

SIMPLIFICATION

SET 3

Q.01. On comparing the following two numeric expressions, we find that _____.

$$[(2\frac{7}{9})^{\frac{3}{2}}]^{\frac{3}{5}} \text{ and } [(1\frac{2}{3})^5]^{\frac{3}{5}}$$

- (a) both the expressions are equal
(b) the first expression is smaller than the second
(c) the first expression is larger than the second
(d) the given two expressions cannot be compared

Q.02. If $x = (\frac{3}{2})^2 \times (\frac{2}{3})^{-4}$, find the value of x^{-2} .

- (a) $(\frac{3}{2})^6$ (b) $(\frac{3}{2})^{12}$
(c) $(\frac{2}{3})^{12}$ (d) $(\frac{2}{3})^6$

Q.03. If

$$\sqrt[3]{0.08 \times 0.8 \times p} = 0.008 \times 0.8 \times \sqrt[3]{q},$$

then find the value of $\frac{p}{q}$.

- (a) $8^3 \times 10^{-8}$ (b) $8^3 \times 10^{-9}$
(c) $8^4 \times 10^{-9}$ (d) $8^4 \times 10^{-8}$

Q.04. Simplify :

$$(\sqrt[3]{\sqrt[3]{343}})^3 \times (\sqrt[3]{\sqrt[3]{216}})^3$$

- (a) 56 (b) 42 (c) $\sqrt[3]{42}$ (d) 42^3

Q.05. Simplify $x^9 \times x^5 \times x^{-4} \times x^0 \times x^{-6}$

- (a) x^4 (b) x^{-4} (c) x^{-6} (d) x^6

Q.06. The value of

$$\frac{\sqrt[3]{-2744} \times \sqrt[3]{-216}}{\sqrt[3]{\frac{64}{729}}} \text{ is}$$

- (a) 164 (b) 152 (c) 189 (d) 156

Q.07. What is the value of 'a' in the below equation ?

$$\{(5 \times 5 \times 5 \times 5 \times 5 \times 5) \times (5 \times 5 \times 5 \times 5 \times 5)\} \div (5 \times 5) = (625)^a$$

- (a) 14 (b) 17 (c) 16 (d) 15

Q.08. Select the number that will come in place of the question mark(?) in the mathematical statement.

$$(0.064)^{123} \div (0.16)^{47} \times (0.4)^{34} \times (0.4)^{29} = (0.4)^?$$

- (a) 350 (b) 320 (c) 338 (d) 341

Q.09. Simplify $[7\frac{1}{2} \div \{1\frac{1}{4} - \frac{1}{2}(2\frac{1}{2} - \frac{1}{4} - \frac{1}{6})\}] \div (\frac{1}{2} \text{ of } 8\frac{1}{3}) = ?$

- (a) 26 (b) 36 (c) 43.2 (d) 23.6

Q.10. Simplify :

$$3p - [3p - p + q - \{3p - (p - q - p)\}]$$

- (a) $3p - q$ (b) $p + 2q$
(c) $3p + 2q$ (d) $2p + 2q$

Q.11. If $\frac{2x+3}{18 \div 3 \times 4 + 2} = \frac{12 \times 3 + 5}{2 \times 4 + 2}$,

then what is the value of x ?

- (a) 132.25 (b) 131.75 (c) 134.35 (d) 120.5

Q.12. If $A = \frac{2}{3} + \frac{4}{7}$ of $\frac{14}{12}$ and $B = \frac{5}{12}$

of $\frac{7}{15} \times \frac{36}{21}$, then what is the value of A + B ?

- (a) $\frac{5}{3}$ (b) 2 (c) $\frac{4}{3}$ (d) $\frac{2}{3}$

Q.13. What is the value of

$$\frac{\frac{5}{6} - \frac{5}{11} \text{ of } [\frac{3}{7} + \frac{8}{12} \div \frac{6}{24} + \frac{5}{7}]}{\frac{3}{12} \times \frac{6}{5}} ?$$

- (a) $-\frac{1891}{691}$ (b) $\frac{2075}{693}$
(c) $-\frac{1894}{891}$ (d) $-\frac{2075}{693}$

Q.14. If $P = 7 \times 4 - 3$, $Q = 8 - 4 + 2$, and $R = 14 \div 21 \times 9$, then what is the value of $P \times Q + R$?

- (a) 96 (b) 112 (c) 108 (d) 106

Q.15. Find the value of the following expression.

$$[\frac{5}{8} - \{\frac{3}{8} - (\frac{5}{8} - \frac{3}{8})\}] \text{ of } 8.8 - 1.2$$

$$4\frac{1}{6} \div 2.5 \times 2 \div \frac{1}{6} \text{ of } 60 + (\frac{3}{4} - \frac{3}{8})$$

- (a) $5\frac{22}{43}$ (b) $3\frac{23}{67}$ (c) $4\frac{44}{85}$ (d) $4\frac{4}{5}$

Q.16. Solve the following.

$$24 \div \frac{3}{8} \text{ of } (8 + 2 \times \overline{7-3}) + [\frac{2}{11} \div \frac{4}{55} - \{\frac{5}{8} + \frac{6}{16}\}]$$

$$32 \div 15 - 7 + 75 \div (6 + 15 \div 3 + 4)$$

- (a) $\frac{23}{27}$ (b) $\frac{9}{2}$ (c) $\frac{11}{18}$ (d) $\frac{15}{19}$

Q.17. Simplify the following expression.

$$9\frac{1}{5} \div \{(16\frac{1}{5} \div 12\frac{2}{3} - 9\frac{2}{3}) +$$

$$17\frac{1}{2} \text{ of } 3\frac{1}{21}\}$$

- (a) $\frac{139}{881}$ (b) $\frac{137}{881}$ (c) $\frac{140}{881}$ (d) $\frac{138}{881}$

Q.18. The value of

$$\frac{1}{4} + \frac{[(20.35)^2 - (8.35)^2] \times 0.0175}{(1.05)^2 + (1.05)(27.65)} \text{ is:}$$

- (a) $\frac{3}{20}$ (b) $\frac{3}{10}$ (c) $\frac{7}{20}$ (d) $\frac{9}{20}$

Q.19. The simplified value of

$$\frac{119 - [48 \div 6 - 7\{5 \times 12 \div 3 - (15 - 3 - 8)\}]}{24 \div 3 - 9 \text{ of } 3 + (52 - 8 - 4) \div 6}$$

is:

- (a) $-\frac{111}{11}$ (b) $-\frac{127}{27}$ (c) $\frac{127}{43}$ (d) $\frac{111}{11}$

Q.20. If the following interchanges are made in signs and numbers, which equation would be correct ?

Interchanges: Signs: + and - Numbers: 15 and 5

- (a) $4 \times 30 - 15 \div 5 + 12 = 21$
(b) $12 \times 30 - 15 \div 5 + 4 = 37$
(c) $4 \times 30 - 15 \div 5 + 12 = 27$
(d) $4 \times 30 - 5 \div 15 + 14 = 21$

Q.21. If '+' means ' $\div$ ', ' $\div$ ' means '+', '>' means '<' and '<' means '>' then the

value of $\frac{42 - 12 \times 3 + 8 \div 2 + 15}{8 \times 2 - 4 + 9 \div 3}$ is:

- (a) $-\frac{5}{3}$ (b) $-\frac{15}{19}$ (c) $\frac{15}{19}$ (d) $\frac{5}{3}$

Q.22. If $x = \frac{1}{12.13} + \frac{1}{13.14} + \frac{1}{14.15}$

$$..... + \frac{1}{23.24}, y = \frac{1}{36.37} + \frac{1}{37.38} +$$

$$\frac{1}{38.39} + \frac{1}{71.72} \text{ then } \frac{x}{y} \text{ is equal to :}$$

- (a) $\frac{1}{3}$ (b) $\frac{1}{24}$ (c) $\frac{1}{72}$ (d) 3

Q.23. The simplified value of

$$\frac{0.01404}{24^2 + 6^2 - 144} \text{ is:}$$

- (a) 3×10^{-5} (b) 6×10^{-5}
(c) 2.4×10^{-4} (d) 3×10^{-4}

Q.24. The simplified value of

$$\frac{1.0025 + 6.25 \times 10^{-6}}{0.0025 + 0.95} \text{ is:}$$

- (a) 1.0025 (b) 1.0525
(c) 1.0005 (d) 1.0505

Q.25. What is the value of

$$\frac{\frac{5}{2} \div \frac{2}{5} \times \frac{1}{5} - \frac{1}{5} \div \frac{3}{5} \times \frac{5}{2}}{\frac{2}{5} \div \frac{1}{5} - \frac{1}{2} \times \frac{5}{2} \div \frac{25}{4} + \frac{1}{2} \div \frac{2}{5} \times \frac{1}{5}}$$

- (a) $\frac{31}{127}$ (b) $\frac{32}{127}$ (c) $\frac{28}{125}$ (d) $\frac{25}{123}$

Q.26. If $A = \frac{(36 \div 18 \times 5)}{(36 \div (3 \times 4))}$ and

$B = \frac{(5 \div 15 \times 5 + 6 \times 2 - 5)}{(15 \div 5 \times 25 - 2 \times 3 + 6)}$ then what is the value of (A + B) ?

- (a) $\frac{776}{225}$ (b) $\frac{424}{225}$ (c) $\frac{772}{225}$ (d) $\frac{401}{225}$

Q.27. If $\frac{1}{4}$ of $\frac{4}{5} + \frac{9}{2}$ of $\frac{17}{3} - \frac{3}{10} = p$, then what is the value of p ?

- (a) $\frac{127}{5}$ (b) $\frac{177}{5}$ (c) $\frac{255}{2}$ (d) $\frac{51}{2}$

Q.28. If $A = \frac{18 \div 9 \times 4}{15 \div 3 \times 5}$ and

$B = \frac{18 \div 36 \times 3 + 5 \times 1 - 6}{18 \div 6 \times 20 - 3 \times 4 + 12}$ then what is the value of $(A + B)$?

- (a) $\frac{137}{600}$ (b) $\frac{23}{100}$ (c) $\frac{187}{600}$ (d) $\frac{197}{600}$

Q.29. Simplify the following. $81^{\frac{3}{4}} + \{(20 \div 5 \text{ of } 3 \times 6) + \{(8 \div 24 \text{ of } 3) \times 4\} - 10 \div 5\} - (\frac{1}{32})^{-\frac{2}{5}}$

- (a) $24\frac{1}{4}$ (b) $21\frac{1}{9}$ (c) $27\frac{4}{5}$ (d) $29\frac{4}{9}$

Q.30. Solve the following equation. $(7^3 - 7^2 + 6) \times (6^3 - 2^3 - 3^2 + 1) \div (21^2 + 20^2 - 41) =$ _____

- (a) 75 (b) 40 (c) 50 (d) 25

Q.31. What is the positive value of the following expression?

$$\sqrt{36 \div 15 \text{ of } 2 \text{ of } [25 \times 4 \div 4 \text{ of } \{29 + (12 - 11)\}]}$$

- (a) $1\frac{5}{6}$ (b) $1\frac{1}{5}$ (c) $2\frac{4}{5}$ (d) $2\frac{3}{5}$

Q.32. Find the value of:

$$\frac{49}{63} \div \frac{21}{9} - \frac{85}{510} + \frac{88}{132} \div \frac{66}{11} - \frac{1}{7} \div \frac{33}{21} + \frac{26}{143} - \frac{1}{209} \div \frac{1}{19} ?$$

- (a) $\frac{55}{36}$ (b) $\frac{55}{39}$ (c) $\frac{13}{46}$ (d) $\frac{44}{13}$

Q.33. What is the value of $\frac{1.2 \times 1.2 + 0.2 \times 0.2 + 1.2 \times 0.4}{1.8 \times 1.8 + 0.4 \times 0.4 - 1.8 \times 0.8}$?

- (a) 0.1 (b) 0.25 (c) 0.2 (d) 1

Q.34. What is the value of A .

$$\text{if } 2 + 4 + (A + 6) \times 2 = \frac{16}{7} ?$$

- (a) 22 (b) 20 (c) 23 (d) 21

Q.35. What is the value of x .

$$\text{if } (\frac{1}{2}) \text{ of } 20 + (\frac{1}{4}) \text{ of } 28 - (\frac{1}{3}) \text{ of } x = 10 ?$$

- (a) 21 (b) 7 (c) 28 (d) 10

Q.36. Find the value of

$$\sqrt{[(13^2 - 8^2) \times 5 + 4]} :$$

- (a) 22 (b) 24 (c) 26 (d) 23

Q.37. Find the value of

$$\sqrt{[(7^2 + 6^3 + 9^2) \div 2 + 52]}$$

- (a) 14 (b) 15 (c) 16 (d) 13

Q.38. The value of $74 + 88 \div 8 \times 10$ of $5 - 90 \div 15 + \{44 - (108 \div 12)\}$ is:

- (a) 700 (b) 653 (c) 770 (d) 605

Q.39. $48 \div 4 + A \times 2 - 8 = 14$, find out the value of A .

- (a) 10 (b) 5 (c) 14 (d) 8

Q.40. What is the value of

$$\frac{2.5 \times 2.5 + 0.3 \times 0.3 + 5 \times 0.3}{1.6 \times 1.6 + 0.6 \times 0.6 - 1.2 \times 1.6} ?$$

- (a) 7.64 (b) 8.64 (c) 2.8 (d) 7.84

RECURRING

Q.41. The value of $22.\overline{4} + 11.\overline{567} - 33.\overline{59}$ is

- (a) $0.\overline{32}$ (b) $0.\overline{412}$ (c) $0.\overline{31}$ (d) $0.\overline{412}$

Q.42. The value of

$$(2.\overline{4} \times 0.\overline{6} \times 3 \times 0.\overline{16}) \times [0.\overline{27} \times (0.\overline{83} \div 0.\overline{16})] \text{ is:}$$

- (a) $0.\overline{814}$ (b) $0.\overline{11}$ (c) $1.\overline{1}$ (d) $1.\overline{36}$

Q.43. The value of $0.\overline{56} - 0.\overline{723} + 0.\overline{39} \times 0.\overline{7}$ is:

- (a) $0.\overline{154}$ (b) $0.\overline{154}$
(c) $0.\overline{158}$ (d) $0.\overline{158}$

Q.44. If $A = 0.\overline{312}$, $B = 0.\overline{415}$ & $C = 0.\overline{309}$, then what is the value of $A + B + C$?

- (a) $\frac{1141}{1100}$ (b) $\frac{1097}{1100}$ (c) $\frac{1211}{1100}$ (d) $\frac{1043}{1100}$

MISCELLANEOUS

Q.45. The simplified form of the expression

$$\frac{(x^2 - 9) \times (9x^2 - 1)}{(x - \frac{1}{3}) \times (1 - \frac{x}{3})} \times \frac{9}{(x + \frac{1}{3})(1 + \frac{x}{3})}$$

is _____

- (a) -729 (b) -81 (c) 729 (d) 81

Q.46. Simplify the following expression.

$$0.01 + 0.68 \times 0.68 + 0.22 \times 0.22 + 0.044 + 0.136 + 0.44 \times 0.68$$

- (a) 0.69 (b) 1 (c) 0.8 (d) 0.9

Q.47. What is the value of the given Expression?

$$\frac{4^{a+4} - 5 \times 4^{a+2}}{15 \times 4^a - 2^2 \times 4^a}$$

- (a) 16 (b) 64 (c) 20 (d) 24

Q.48. What is the value of $(2^2 + 1)(2^4 + 1)$

$$(2^8 + 1)(2^{16} + 1) \dots (2^{128} + 1) ?$$

- (a) $\frac{2^{256} - 1}{2}$ (b) $\frac{2^{256} - 1}{3}$
(c) $2^{256} - 1$ (d) $\frac{2^{256} - 1}{4}$

Q.49. Which of the following statement is true?

$$I. \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{110} < \frac{5}{6}$$

$$II. \frac{1}{3} + \frac{1}{15} + \frac{1}{35} + \dots + \frac{1}{143} > \frac{7}{13}$$

- (a) Only I (b) Both I and II
(c) Only II (d) Neither I nor II

Q.50. Simplify the following.

$$\frac{0.01 \times 0.01 \times 0.01 + 0.003 \times 0.003 \times 0.003}{0.05 \times 0.05 - 0.015 \times 0.05 + 0.015 \times 0.015}$$

- (a) $\frac{13}{25} \times 10^3$ (b) $\frac{13}{15} \times 10^{-3}$
(c) $\frac{13}{15} \times 10^3$ (d) $\frac{13}{25} \times 10^{-3}$

Q.51. What is the correct descending order of the following ratios?

$$\frac{17}{30}, \frac{7}{15}, \frac{27}{50}, \frac{11}{20}$$

- (a) $\frac{11}{20} > \frac{17}{30} > \frac{27}{50} > \frac{7}{15}$
(b) $\frac{17}{30} > \frac{11}{20} > \frac{27}{50} > \frac{7}{15}$
(c) $\frac{17}{30} > \frac{27}{50} > \frac{11}{20} > \frac{7}{15}$
(d) $\frac{7}{15} > \frac{11}{20} > \frac{27}{50} > \frac{17}{30}$

Q.52. Find the value of $0.5 + 0.55 + 0.555 + 0.5555$:

- (a) 1.4405 (b) 1.0555
(c) 1.5555 (d) 2.1605

Q.53. What is the difference of the largest and smallest of the given fractions?

$$\frac{5}{11}, \frac{5}{7}, \frac{3}{8}, \frac{6}{13}$$

- (a) $\frac{17}{56}$ (b) $\frac{19}{56}$ (c) $\frac{1}{7}$ (d) $\frac{23}{56}$

Q.54. What is the value of

$$99\frac{11}{99} + 99\frac{13}{99} + 99\frac{15}{99} + \dots + 99\frac{67}{99} ?$$

- (a) $\frac{94220}{33}$ (b) $\frac{95120}{33}$
(c) $\frac{97120}{33}$ (d) $\frac{96220}{33}$

Q.55. What is the simplified value of

$$\frac{(x + y + z)(xy + yz + zx) - xyz}{(x + y)(y + z)(z + x)} ?$$

- (a) y (b) z (c) 1 (d) x

Q.56. Simplify: $6\frac{4}{5} \div [4\frac{1}{4} \div \{5\frac{1}{3} \times \frac{3}{8} + (3\frac{1}{2} + \frac{2}{3}) \times \frac{4}{5}\}]$

- (a) $8\frac{8}{15}$ (b) $7\frac{8}{25}$ (c) $8\frac{8}{25}$ (d) $7\frac{8}{15}$

Q.57. Simplify the given expression.

$$\frac{(0.25)^4 + 2 \times (0.25)^2 + 1 - (0.25)^2}{(0.25)^2 + 0.25 + 1}$$

- (a) 0.6755 (b) 0.9025
(c) 0.8125 (d) 0.7835

Q.58. What is the value of

$$\sqrt{0.48 \times 0.48 + 0.25 \times 0.25 + 0.5 \times 0.48}$$

- (a) 0.63 (b) 0.93 (c) 0.83 (d) 0.73